Sequence Models as Proxies for Flow-Matching-Based Alignment

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In one sentence. We replace the Markovian velocity field and ODE solver of flow-matching alignment with a sequence model (**GRUFlow** or **TFFlow**) that predicts every trajectory state directly. Its memory of the path lets it tell crossing trajectories apart, which a memoryless velocity field cannot do.

50× lower transport error than FMA on crossing paths (XOR task)

+2.3% 1-shot accuracy over FMA, difficult-dataset average

39% of FMA's parameters (GRUFlow 6.5 M vs. 16.65 M)

+22.5% on frozen zero-shot CLIP features (TFFlow, 16-shot)

Motivation

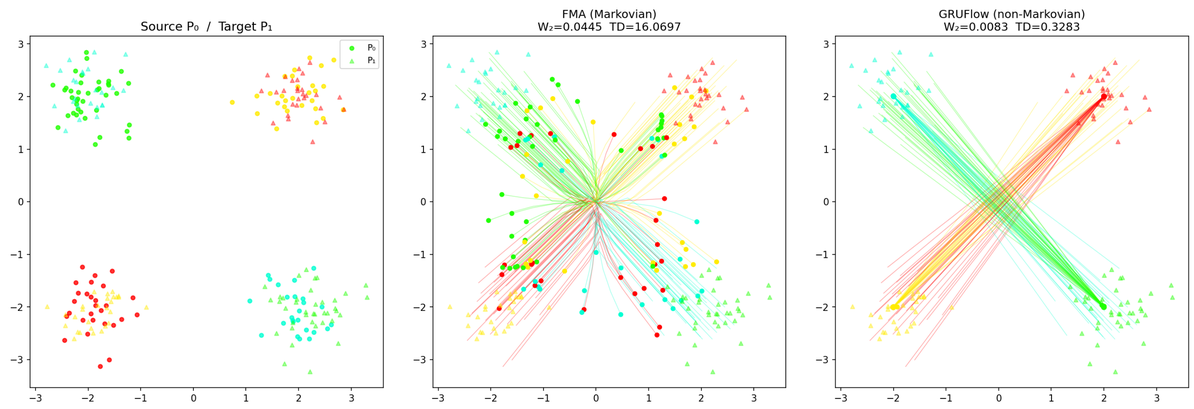
Flow-Matching Adaptation (FMA) [1] aligns a CLIP image feature x_0 with its class text prototype x_1 by integrating a learned velocity field,

$$\frac{dx_t}{dt} = v_\theta(x_t, t), \qquad x_0 = f_{\text{img}}(I).$$

This has three practical problems:

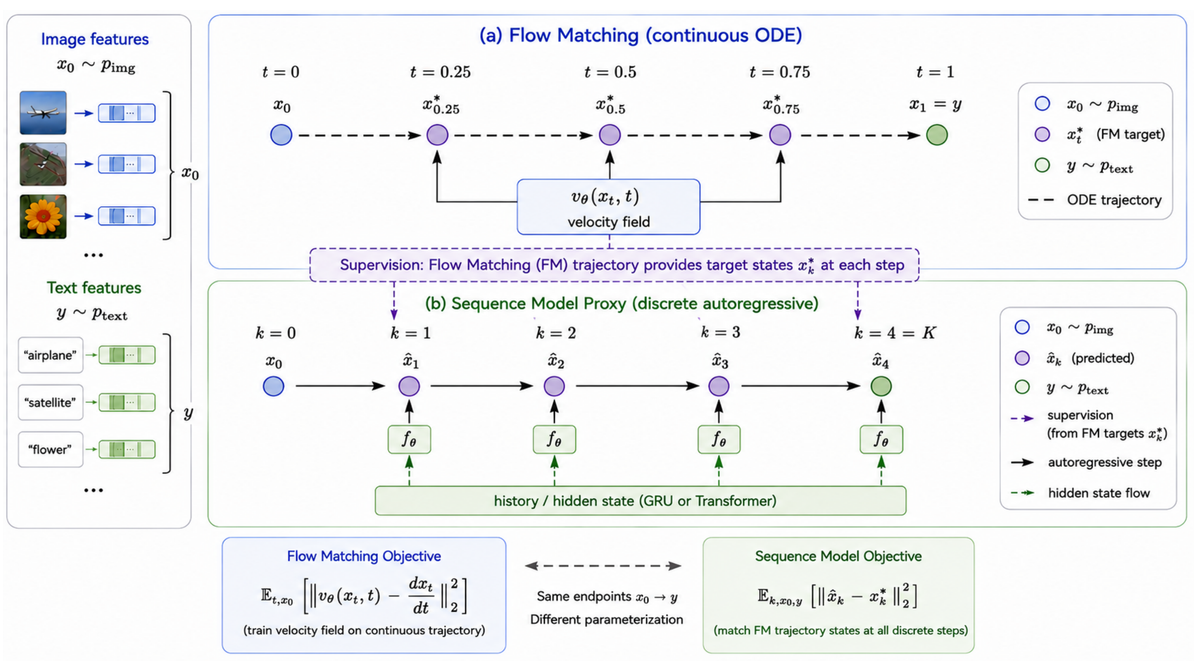
1. **Solver tuning.** Step size, horizon and early stopping all need tuning.
2. **No memory.** v_θ sees only the current state, so it cannot tell crossing paths apart.
3. **Indirect supervision.** It regresses a sampled velocity, never the states on the path.

Question. Can a sequence model stand in for FM-based alignment, replacing ODE integration with learned autoregressive state prediction?



XOR toy task: four clusters swap diagonals. FMA funnels every path through the centre and mixes up targets; GRUFlow's hidden state keeps each cluster on its own path.

Method: predict states, not velocities



Split the path into $K{=}10$ steps, $t_k = k/K$. Each step is supervised by the closed-form interpolant, with no ODE solve:

$$x^\star_{t_k} = (1 - t_k)\, x_0 + t_k\, x_1 + \gamma\sqrt{t_k(1 - t_k)}\,\varepsilon, \qquad \mathcal{L}(\theta) = \frac{1}{K}\sum_{k=1}^{K} w_k \left\|\hat{x}_{t_k} - x^\star_{t_k}\right\|^2$$

GRUFlow — 6.5 M params

- GRU hidden state $\mathbf{h}_k$ carries the trajectory memory
- 3 AdaLN decoder blocks conditioned on t_k
- Residual update $\hat{x}_{t_k} = \hat{x}_{t_{k-1}} + W_{\text{out}}\tilde{h}_k$
- $\mathcal{O}(K)$ rollout

TFFlow — 9.82 M params

- Causal self-attention over the history $[x_0, \hat{x}_{t_1}, \ldots, \hat{x}_{t_{k-1}}]$
- 4 blocks, 8 heads, AdaLN on t_k
- Last token gives the same residual update
- $\mathcal{O}(K^2)$ rollout

Closed-loop training feeds each prediction back as the next input, exactly as at test time (no exposure bias). **Inference:** unroll from x_0, ℓ_2-normalise $\hat{x}_{t_K}$, pick the nearest text prototype. No ODE solver.

Synthetic transport: does memory matter?

Six 2-D tasks with known ground-truth pairings, mean of 5 seeds. FMA (530 K params) vs. GRUFlow (397 K). W_2 measures distribution match and is blind to swapped labels; **TD** measures error to each sample's *true* target.

	$W_2\downarrow$		TD↓		
Task	FMA	GRUFlow	FMA	GRUFlow	TD ratio
Figure-8	0.0335	0.0334	0.0051	0.0048	1.1×
Multi-Ring	0.0352	0.0350	0.0724	0.0657	1.1×
Spiral	0.0276	0.0269	0.1079	0.0715	1.5×
Swiss Roll	0.0226	0.0214	0.0741	0.0050	**14.8**×
Gaussian Swap	0.0526	0.0158	9.7235	0.4796	**20.3**×
XOR	0.0451	0.0079	15.6368	0.3074	**50.9**×

On simple paths (Figure-8, Multi-Ring) the two tie. Where paths cross, branch or follow a manifold, TD drops 14.8–50.9×. W_2 hides the failure: on Gaussian Swap FMA reaches $W_2{=}0.053$ while its TD is 9.72.

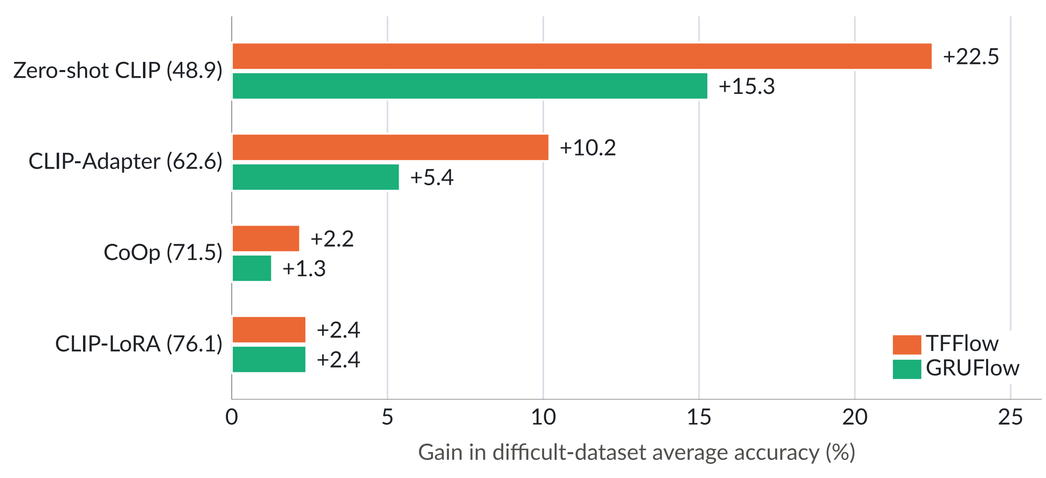
Few-shot classification: 11 datasets

CLIP ViT-B/16. Averages over 5 difficult (Aircraft, EuroSAT, DTD, SUN397, Cars) and 6 easy datasets. All refiners run on CLIP-LoRA features.

		1-shot		4-shot		16-shot	
Method	Params	Diff.	Easy	Diff.	Easy	Diff.	Easy
CLIP-LoRA [2]	–	58.7	83.0	67.3	85.6	76.1	88.2
+ FMA [1]	16.65 M	59.4	83.8	69.0	86.5	77.8	88.8
+ TFFlow (ours)	9.82 M	61.1	84.2	**70.4**	**86.8**	**78.5**	88.9
+ GRUFlow (ours)	6.5 M	**61.7**	**84.4**	70.1	**86.8**	**78.5**	**89.1**

- Gain over FMA on difficult datasets shrinks as shots grow: +2.3, +1.1, +0.7 (GRUFlow). Fewer labels means more ambiguity for memory to resolve.
- Largest single gain: EuroSAT 1-shot, 73.0 → 82.3 over FMA.
- On easy datasets all refiners stay within 0.6 points of FMA.

Works on any PEFT backbone (16-shot)



Backbone's own difficult-dataset average in brackets. Easy-dataset gains are smaller but follow the same trend (zero-shot CLIP: +7.0 GRUFlow, +9.5 TFFlow).

Gains are largest on the weakest backbones. TFFlow's full causal context helps most there; on CLIP-LoRA the two models tie.

Takeaways

1. **Memory does the work.** Trajectory context resolves crossing paths that a memoryless velocity field gets wrong.
2. **No ODE needed.** Discrete, directly supervised refinement steps carry the multi-step alignment gains.
3. **Smaller.** Match or beat FMA with 39–59% of its parameters, with the biggest gains at 1-shot, on difficult datasets and on weak backbones.
4. **Which model?** GRUFlow is the efficient default; TFFlow is best on weak feature extractors.

[1] Z. Jiang, Y. Wang, L. Chen. Exploring Cross-Modal Flows for Few-Shot Learning. ICLR 2026.
[2] M. Zanella, I. Ben Ayed. Low-Rank Few-Shot Adaptation of Vision-Language Models. CVPR 2024.

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